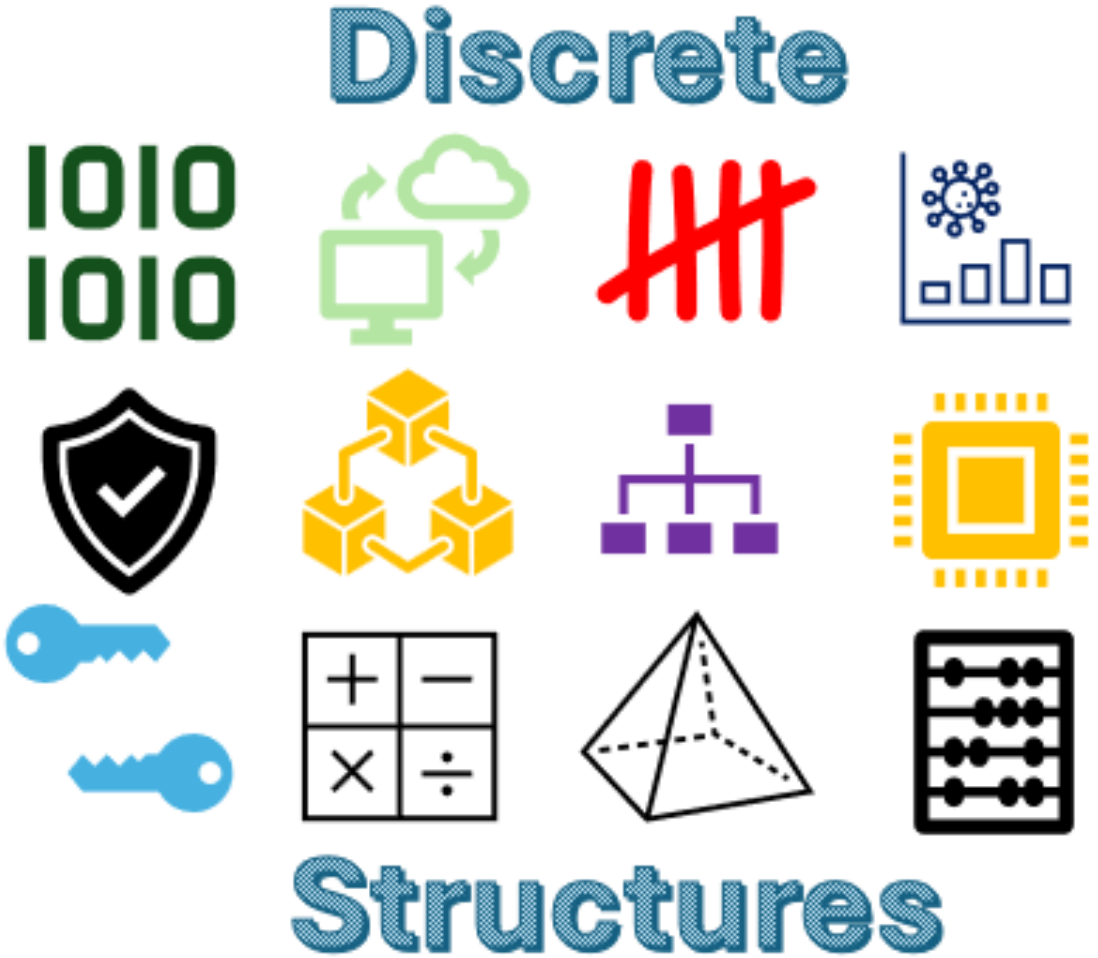
COMP 2310: Discrete Structures
Fall 2026

Chapter 3: Counting



Pigeonhole principle

- Fundamental and powerful counting technique
- **Theorem:** If *n* pigeons are assigned to *m* pigeonholes, and *m* < *n*, then at least one pigeonhole contains two or more pigeons.
  - If every pigeonhole contains at most one pigeon, at most m pigeons can be assigned.
  - Since m<n, this is a contradiction.
  - **This contradiction leads to a conclusion that:** at least one pigeonhole must contain two or more pigeons. BUT does not provide information about that specific PIGEONHOLE



https://en.wikipedia.org/wiki/Pigeonhole_principle

Pigeonhole principle: example 1

- If 8 people are chosen, at least two of them share the same day of the week they were born on.
  - 7 pigeonholes (days of the week), 8 pigeons (people)
  - By the Pigeonhole Principle, at least two people share the same day of birth
- Provides an existence proof
  - Guarantees that there exists at least two people (“objects”) with this characteristics (born on the same day)
  - But does not show how to find those people (“objects”)

Pigeonhole principle: example 1

- Show that if any 5 numbers are chosen from the set {1, 2, 3, ..., 8}, two of them will sum to 9
  - **Pigeons:** number of chosen objects: the **five** chosen number from {1, 2, 3, ..., 8}
  - **Pigeonholes:** numbers of pairs that will sum to 9:
    - The **four** pairs of numbers that sum to 9: {1,8}, {2,7}, {3,6}, and {4,5}
  - Since there are 4 pigeonholes and 5 chosen numbers (select one number from each pair→4 numbers, you must select another number to make to choose 5 numbers
    - the 5th number must come from a pigeonhole where a number has already been chosen

Pigeonhole principle: example 2

- Show that if 11 numbers are chosen from {1, 2, ..., 20}, one number will be a multiple of another.
  - Pigeons: 11 numbers
  - Pigeonholes: 10 or a smaller number of pairs
    - *Identify the odd part*
    - Every positive integer can be expressed as $n = 2^k.m$ ($m \geq 0$)
      - m = odd part of n
      - k = number of times k divides n
      - E.g. $12 = 2^2.3$ (k = 2, m = 3) ➔ 3 is the odd part
      - If 11 numbers are chosen from the set, then then two of them must have the same odd part
      - Follows the Pigeonhole principle: 11 numbers but 10 odd numbers
      - **How does it ensure that one of the chosen numbers in a pair is a multiple of another?**
        - If two numbers have the same odd part, then ONE number is the multiple of the other.
          - How to find out?

Pigeonhole principle: example 2

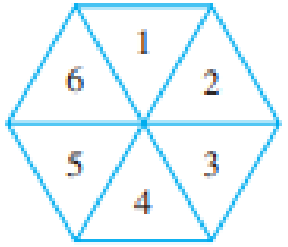
- Let *n*1 and *n*2 be two chosen numbers with the same odd part.
- There are only **10 possible odd parts** among $1, \ldots, 20$: $1,3,5,7,9,11,13,15,17,19$
- We can express these numbers as:
  - $n1 = 2^{k1}.m$
  - $n2 = 2^{k2}.m$
  - If $k1 \geq k2$, then n1 is a multiple of n2
  - If $k2 \geq k1$, then n2 is a multiple of n1

Pigeonhole principle: example 4

- Consider the region shown below. It is bounded by a regular hexagon whose sides are of length 1 unit. Show that if any seven points are chosen in this region, then two of them must be no farther apart than 1 unit.



- Solve:
  - Identify a way of solving the problem so that the number of objects chosen (Pigeons) are more than the number of Pigeonholes
  - Consider divide the region in six (6 < 7) equilateral triangle.
  - Therefore, 7 Pigeons (points) and 6 Pigeonholes (Triangles)



  - According to the Pigeonhole Principle, if you have more points (7) than you have triangles (6), then at least two of those points must be located within the same triangle.

Application Pigeonhole principle

- Database systems/data analysis
  - More records than the unique identifier
- Scheduling problem
  - More entities than slots
- Network security
  - More devices than IP address
- Conflict resolution

Extended Pigeonhole principle

- Note that if there are *m* pigeonholes and more than 2*m* pigeons, three or more pigeons will have to be assigned to at least one of the pigeonholes.
- If *n* pigeons are assigned to *m* pigeonholes, then one of the pigeonholes must contain at least $\lfloor (n-1)/m \rfloor + 1$ pigeons.

Example 7 Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.

Solution

Let the pages be the pigeons and the dictionaries the pigeonholes. Assign each page to the dictionary in which it appears. Then by the extended pigeonhole principle, one dictionary must contain at least $\lfloor 61{,}326/30 \rfloor + 1$ or 2045 pages. ◆

Exercise

- How many friends must you have to guarantee at least five of them will have birthdays in the same month?
  - Hints: use the extended Pigeonhole principle.
  - Pigeonhole m = ?
  - Number of Pigeons n (how many friends) = ?

  - Hints: use the extended Pigeonhole principle.
  - Pigeonhole m = 12
  - $\lfloor (n-1)/m \rfloor + 1$ = 5

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Elements of probability

- Counting techniques are applied in probability
- **Probabilistic experiments**: Outcomes vary with each repetition (e.g., coin tosses).
- A sample space is a set of all possible outcomes of an experiment.
  - Tossing a nickel and a quarter
  - Possible outcomes:
    - Number of heads: {0, 1, 2}
    - Sequence of heads/tails: {HH, HT, TH, TT}
    - Coins match (M) or do not match (N): {M, N}

Elements of probability: number of outcomes

- Determine the sample space for an experiment consisting of tossing a six-sided die twice and recording the sequence of numbers showing on the top face of the die after each toss.
  - 6 possible outcomes in the first toss (1, 2, 3, 4, 5, 6)
  - 6 possible outcomes in the second toss (1, 2, 3, 4, 5, 6)
  - Total possible outcomes = 6 x 6 = 36

Elements of probability: events

- An event is a statement about the outcome of an experiment (either true or false).
- Events are subsets of the sample space
- **Example**: Rolling two dice.
  - Event 1: Both dice show less than 3.
  - Event 2: Sum of the dice equals 4.

- **Event A (both dice less than 3):** (1,1), (1,2), (2,1), (2,2), so P(A) = 4/36 = 1/9
- **Event B (sum equals 4):** (1,3), (2,2), (3,1), so P(B) = 3/36 = 1/12

Elements of probability: Assigning probabilities

- Probabilities reflect the likelihood of events.
- The number p(E) reflects the assessment of the likelihood that the event E will occur.
- If the experiment is performed multiple times, and after n performances, the event E occurs n_E times.
  - Then the fraction $f(E) = n_E/n$, called the frequency of occurrence of *E* in *n* trial

Example 6 Suppose an experiment is performed 2000 times, and the frequency of occurrence f_E of an event E is recorded after 100, 500, 1000, and 2000 trials. Table 3.1 summarizes the results.

TABLE 3.1

Number of Repetitions of the Experiment	n_E	$f_E = n_E/n$
100	48	0.48
500	259	0.518
1000	496	0.496
2000	1002	0.501

  - The probability of an even to occur always ranges between 0 to 1.

Elements of probability: axioms

- P1: $0 \leq p(E) \leq 1$ for any event E.
- P2: $p(A) = 1$ for the entire sample space A, $p(\emptyset) = 0$.
- P3: For mutually exclusive events $E_1, E_2, \ldots\ldots, E_k$

$$\text{P3:} \quad p(E_1 \cup E_2 \cup \cdots \cup E_k) = p(E_1) + p(E_2) + \cdots + p(E_k)$$

  - Consider the experiment of tossing a coin
    - Event E: Heads turns up
    - Event F: Tails turns up
  - Symmetry argument: E and F are equally likely since there is no reason to prefer one outcome over the other. Therefore, we assume P(E) = P(F)
  - Since E and F are mutually exclusive, and A represents the entire sample space (E ∪ F):
    - P(A) = P(E) + P(F)
    - ➔1 (see axiom P1) = P(E) + P(E) = 2P(E) since P(E) = P(F)
    - P(E) = 0.5 = P(F)

Elements of probability: example

Example 8 Suppose that an experiment has a sample space $A = \{1, 2, 3, 4, 5, 6\}$ and that the elementary probabilities have been determined as follows:

$$p_1 = \frac{1}{12}, \quad p_2 = \frac{1}{12}, \quad p_3 = \frac{1}{3}, \quad p_4 = \frac{1}{6}, \quad p_5 = \frac{1}{4}, \quad p_6 = \frac{1}{12}.$$

Let E be the event "The outcome is an even number." Compute $p(E)$.

Solution

Since $E = \{2, 4, 6\}$, we see that

$$p(E) = p_2 + p_4 + p_6 = \frac{1}{12} + \frac{1}{6} + \frac{1}{12} \quad \text{or} \quad \frac{1}{3}.$$

Elements of probability: equally likely outcomes

- let E be an event that contains k outcomes, say $E = \{x_1, x_2, \ldots, x_k\}$

$$p(E) = \underbrace{\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}}_{k \text{ summands}} = \frac{k}{n}.$$

$$p(E) = \frac{|E|}{|A|} = \frac{\text{total number of outcomes in } E}{\text{total number of outcomes}}.$$

- Choose four cards at random from a standard 52-card deck. What is the probability that four kings will be chosen?
  - First find out the ways 4 cards can be chosen from a deck of 52-card (Combination)
  - ${}_{52}C_4 = 270{,}725$
  - Therefore, P(E) = 1 / 270,725

Elements of probability: equally likely outcomes

- A box contains six red balls and four green balls. Four balls are selected at random from the box. What is the probability that two of the selected balls will be red and two will be green?
  - Total number of outcomes = total number of ways 4 balls can be selected at a time from 10 balls
    - Order does not matter, repetition not allowed (Eq 3) = ${}_nC_r$
    - Total outcomes = ${}_{10}C_4$
    - Total number of outcomes for 2 balls being red from 6 red balls P(R)= ${}_6C_2$
    - Total number of outcomes for 2 balls being green from 4 green balls P(G) = ${}_4C_2$
    - Therefore, P(R) + P(G) / ${}_nC_r$
    - $({}_6C_2 + {}_4C_2) / {}_{10}C_4$

Elements of probability: equally likely outcomes

- What is the probability that exactly two coins will land heads up when five fair coins are tossed?
  - Total number of outcomes?
  - Total number of outcomes for 2 coins will land heads?

- Total number of outcomes = Total number of heads and tail arrangements
  - 2^5
  - Total number of outcomes for 2 coins will land heads among 5 coins P(H)= $_5C_2$
  - 10 / 32